RACE Regular Analysis Through Continuous Exercise

Hydrocarbons : RACE-01

Preparation of alkane by hydrogenation of alkene

1. Which of the following statement is/are true for catalytic hydrogenation reaction:

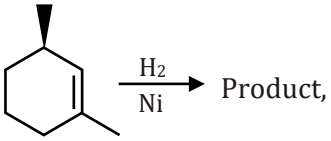
(A) Dihydrogen gas adds to alkenes and alkynes in the presence of finely divided catalysts like Pt, Pd or Ni to form alkanes

(B) The metal adsorbs dihydrogen gas on their surfaces and activate the hydrogen – hydrogen bond.

(C) Pt and Pd catalyse the reaction at room temperature but relatively higher temperature and pressure are required with nickel catalyst.

(D) All of these

CHC090

2. 

Product,

Products are related as

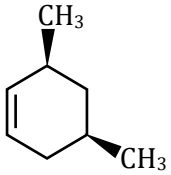
(A) Diastereomers (B) Structural isomers

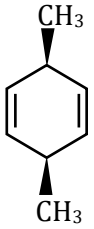
(C) Enantiomers (D) Optically inactive meso structure

CHC288

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3. Which of the following give only meso compound on catalytic reduction with Raney Ni/H_2?

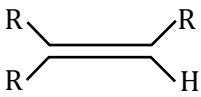
(A) 

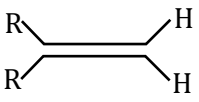
(B) 

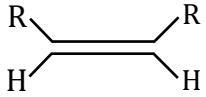
(C) Both (A) & (B) (D) None of these

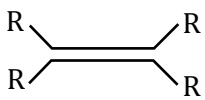
CHC290

4. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition

(A) 

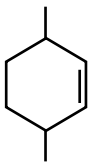
(B) 

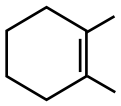
(C) 

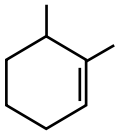
(D) 

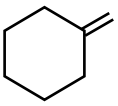
CHC052

5. Decreasing order of heat evolved upon catalytic hydrogenation of given reaction with a H_2/Pd is:

(a) 

(b) 

(c) 

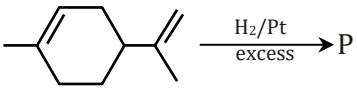
(d) 

(A) b > c > a > d (B) a > d > c > b (C) d > c > a > b (D) c > b > c > d

CHC291

For More Join: @IITJEE_Advanced

6. How many products will be formed for given reaction, including stereochemistry



CHC289

Preparation of alkane by Transfer hydrogenation (NH_2–NH_2+H_2O_2)

7. Which among the following are correct statement about di-imide reduction :
(A) 6-membered cyclic transition state is present.
(B) It is an example of syn addition.
(C) 4-membered cyclic transition state is present.
(D) It reduces C=O, C ≡ N, C=N.

CHC105

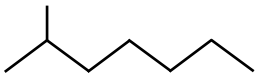
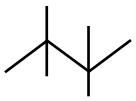
Preparation of alkane by Kolbe's electrolysis

8. Aqueous solution of the following compound on electrolysis gives ethane:
(A) Acetic acid (B) Acetamide
(C) Potassium acetate (D) Ethyl acetate

CHC102

Physical properties and uses of alkane

9. Arrange the following in the decreasing order of their boiling points:

(i) C_9H_{20} (ii) C_8H_{18} (iii)  (iv) 

(A) (i) > (ii) > (iii) > (iv) (B) (iv) > (iii) > (ii) > (i)
(C) (i) > (ii) > (iv) > (iii) (D) (iii) > (iv) > (ii) > (i)

CHC110

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10. **Statement-1:** Boiling point of isopentane is higher than boiling point of 2,2-dimethyl propane.
Statement-2: Surface area of 2, 2-dimethyl propane is lesser than surface area of isopentane.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false
(D) Statement-1 is false, statement-2 is true.

CHC292

Chemical reaction of alkane; Halogenation (Free radical substitution), Reactivity and selectivity principle, Relative rate of reaction (X_2, Alkane), Reed Reaction

11. Which of the following cannot be considered as a step of mechanism in chain reaction of methane with Cl_2 ?
(A) $Cl_2 \longrightarrow 2Cl^{\bullet}$ (B) $CH_4 + Cl^{\bullet} \longrightarrow CH_3Cl + H^{\bullet}$
(C) $Cl^{\bullet} + CH_4 \longrightarrow \dot{C}H_3 + HCl$ (D) $Cl_2 + \dot{C}H_3 \longrightarrow CH_3Cl + Cl^{\bullet}$

CHC118

12. Arrange the following in correct order of reactivity towards $Br_2/h\nu$ -

(i) CH_4 (ii) CH_3CH_3 (iii) $CH_3CH_2CH_3$ (iv) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

(A) i > ii > iii > iv (B) iv > iii > ii > i (C) ii > iii > i > iv (D) iii > ii > iv > i

CHC116

For More Join: @IITJEE_Advanced

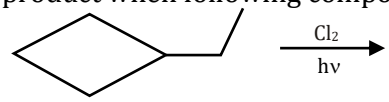
13. $CH_3—\underset{\displaystyle CH_2—CH_3}{\overset{\displaystyle |}{CH}}—CH_2—CH_2—CH_3$

Number of monochlorinated product when following compound undergo reaction with $Cl_2/h\nu$ is :

(A) 10 (B) 15 (C) 18 (D) 20

CHC293

14. Number of monochlorinated product when following compound undergo reaction with $Cl_2/h\nu$ is :



(A) 5 (B) 9 (C) 10 (D) 6

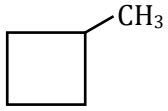
CHC294

15. How many monochlorinated products are obtained on monochlorination of 3-methylpentane:

(A) 3 (B) 5 (C) 6 (D) 8

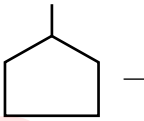
CHC295

16. How many distinct monobromination products, including stereoisomers, can result when the alkane below is heated in the presence of Br_2?



CHC129

17. Consider the following reactions



Total number of monochlorinated product = X (Excluding stereoisomers)



Total number of monochlorinated product = Y (Excluding stereoisomers)

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Then, the value of (X + Y) is

CHC296

Reactions of alkanes; combustion; controlled oxidation [$KMnO_4$, Cu/Heat, Mo_2O_3, $(CH_3COO)_2Mn$]; isomerization; pyrolysis; aromatization and reaction with steam,

18. n–heptane on prolonged heating with a mixture of Al_2O_3 and Cr_2O_3 at 600°C gives mainly :-

(A) Methylcyclohexane (B) Cycloheptane
(C) Methylbenzene (D) Isoheptane

CHC123

19. The final product of complete oxidation of hydrocarbon is:-

(A) Acid (B) Aldehyde (C) $H_2O + CO_2$ (D) Dihydric alcohol

CHC121

20. How many moles of O_2 required for complete combustion of one mole of propane –

CHC125

ANSWER KEY

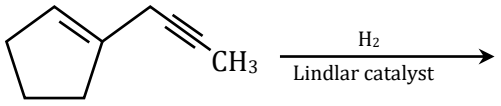
Question	1	2	3	4	5	6	7	8	9	10
Answer	D	A	A	C	B	2	AB	C	A	A
Question	11	12	13	14	15	16	17	18	19	20
Answer	B	B	D	C	D	8	8	C	C	5

For More Join: @IITJEE_Advanced

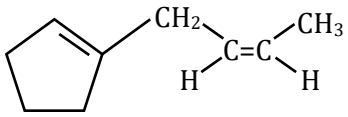
RACE
Regular Analysis Through Continuous Exercise

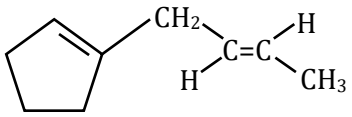
Hydrocarbons : RACE-02

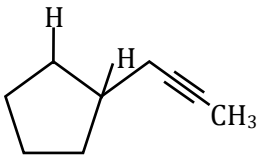
Preparation of alkene by lindlar's catalyst, P-2 catalyst (Ni_2B/H_2), poisoned catalyst

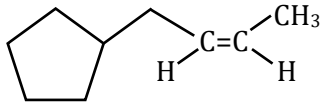
1. 

Product will be :

(A) 

(B) 

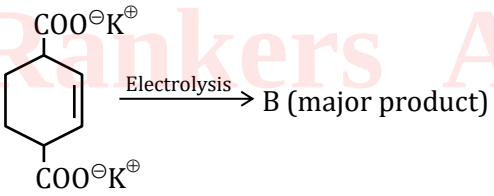
(C) 

(D) 

CHC225

Preparation of alkene by Kolbe's Electrolysis

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2. 

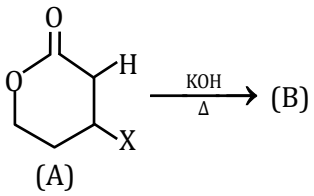
Degree of unsaturation in product B:-

(A) 1 (B) 2 (C) 3 (D) 4

CHC152

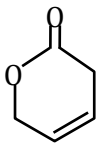
Basic idea of Elimination reactions

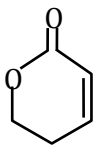
3. When the compound (A) is treated with KOH/Δ, the major product formed is B.

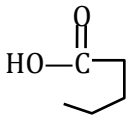


(A)

The product 'B' is :

(A) 

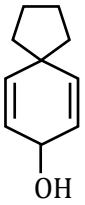
(B) 

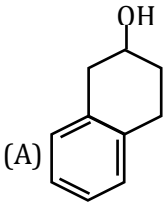
(C) 

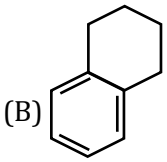
(D) 

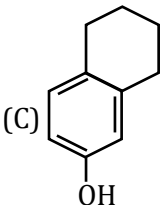
CHC134

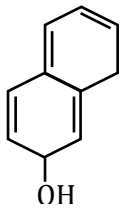
For More Join: @IITJEE_Advanced

4.  $\xrightarrow[\Delta]{H^{\oplus}}$ Product; Product is :

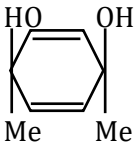
(A) 

(B) 

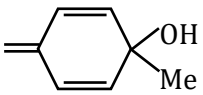
(C) 

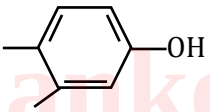
(D) 

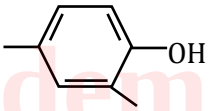
CHC140

5.  $\xrightarrow[\Delta]{H^{\oplus}}$ Major product

(A) 

(B) 

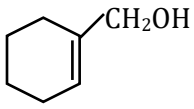
(C) 

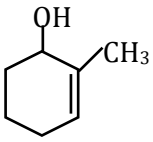
(D) 

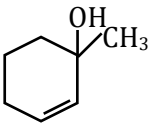
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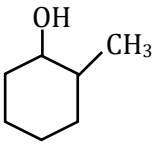
CHC141

6. Identify the correct order of rate of dehydration when given compounds are treated with conc. H_2SO_4 :

(P) 

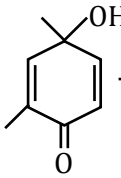
(Q) 

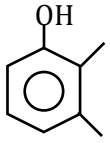
(R) 

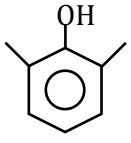
(S) 

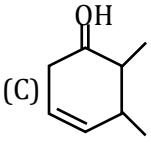
(A) P > Q > R > S (B) Q > P > R > S (C) R > Q > P > S (D) R > Q > S > P

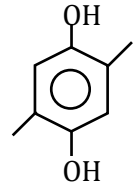
CHC142

7.  $\xrightarrow[\Delta]{H_2SO_4}$ X; X is :

(A) 

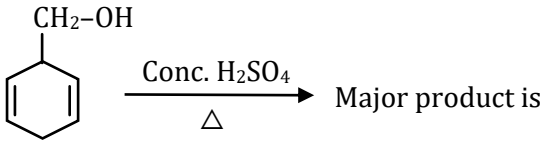
(B) 

(C) 

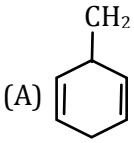
(D) 

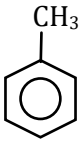
CHC143

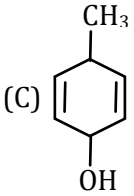
For More Join: @IITJEE_Advanced

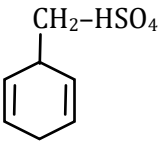
8. 

Major product is

(A) 

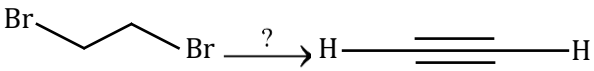
(B) 

(C) 

(D) 

CHC297

9. The reagent(s) for the following conversion,



is / are

(A) aqueous NaOH

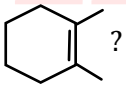
(B) alcoholic KOH followed by $NaNH_2$

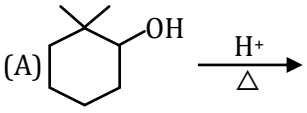
(C) aqueous KOH followed by $NaNH_2$

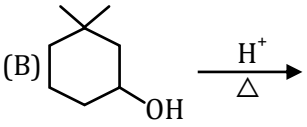
(D) Zn / CH_3OH

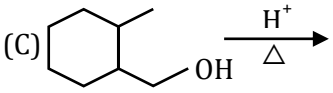
Rankers Academy JEE

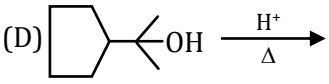
CHC054

10. Which of the following reaction will not produce given alkene as major product  ?

(A) 

(B) 

(C) 

(D) 

CHC300

11. **Statement 1 :** 1,1,1-trideutero-2-propanol reacts with conc. H_2SO_4 at high temperature to give only one alkene, 3,3,3-trideutero propene.

Statement 2 : C–D bond is stronger than C–H bond.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

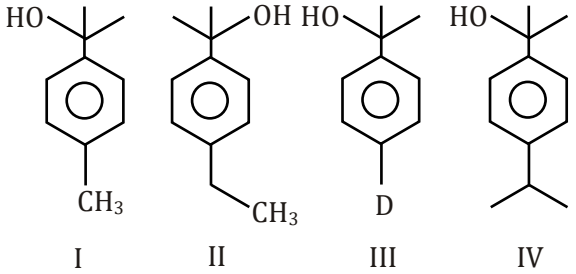
(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true

CHC302

For More Join: @IITJEE_Advanced

12. Correct order for the rate of dehydration for the given compound with H_2SO_4/Δ :

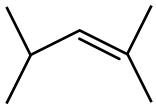


I II III IV

(A) I > II > III > IV (B) IV > II > I > III (C) I > II > IV > III (D) III > IV > II > I

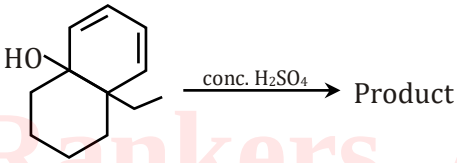
CHC304

13. Number of alcohols which on $H^{\oplus}/\Delta$ gives following alkene as a major product.



CHC301

14. How many monocloroderivatives (with Cl_2, hν) are possible for the major product in the following reaction :



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CHC303

Physical properties and uses of alkene

15. Which one is an correct statement regarding physical property of alkene :-

(A) The first members are gases, the next fourteen are liquids and the higher ones are solid.

(B) Ethene is a colourless gas with faint sweet smell.

(C) They are insoluble in water but fairly soluble in non-polar solvents.

(D) Straight chain alkene have lower boiling point than Isomeric branched chain compounds.

CHC159

16. Which one of the following has the smallest heat of hydrogenation per mole

(A) 1–butene (B) Trans–2–butene

(C) Cis–2–butene (D) Propene

CHC162

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	B	D	C	D	B	B	B
Question	11	12	13	14	15	16				
Answer	D	C	5	11	C	B				

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Hydrocarbons : RACE-03

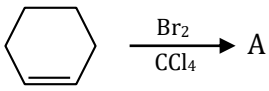
Chemical reaction of alkene; basic idea of addition reaction

1. HBr reacts with $CH_2 = CH - OCH_3$ at room temperature to give

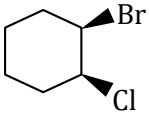
(A) $BrCH_2CHO$ and CH_3OH (B) $BrCH_2-CH_2-OCH_3$

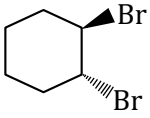
(C) $H_3C-CHBr-OCH_3$ (D) CH_3CHO and CH_3Br

CHC164

2. 

A will have configuration :-

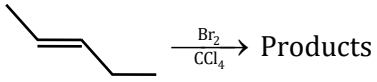
(A) 

(B) 

(C) Both true (D) None is true

CHC168

3. Select the incorrect statement about the product mixture in the following reaction:

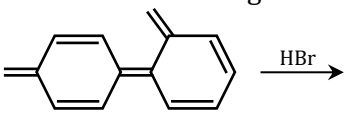


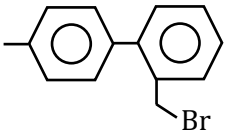
(A) It is optically active (B) It is racemic mixture

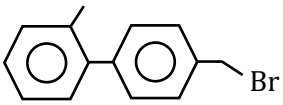
(C) It is a resolvable mixture (D) It is a mixture of erythro compounds

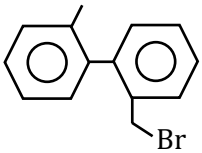
CHC175

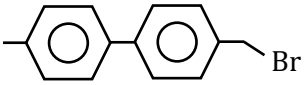
4. Product obtained in given reaction in good yield are :



(A) 

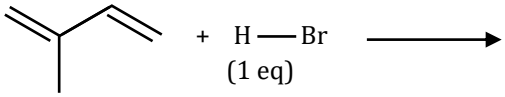
(B) 

(C) 

(D) 

CHC177

5. Correct statement regarding product of reaction:



(A) Kinetic controlled product is a secondary bromide where as thermodynamically controlled product is a tertiary bromide

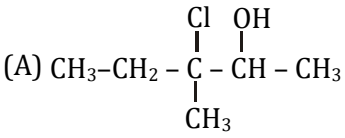
(B) Kinetic controlled product is a tertiary bromide where as thermodynamically controlled product is a primary bromide

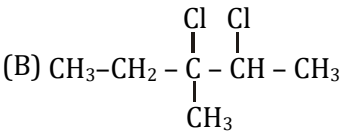
(C) Both kinetic & thermodynamic controlled product is secondary bromide

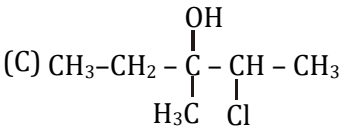
(D) Both kinetic & thermodynamic controlled product is a tertiary bromide **CHC305**

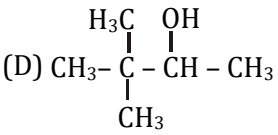
For More Join: @IITJEE_Advanced

6. The predominant product formed, when 3-methyl-2-pentene reacts with HOCl, is :-

(A) 

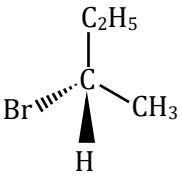
(B) 

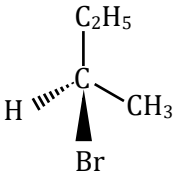
(C) 

(D) 

CHC170

7. The addition of HBr to 1-butene gives a mixture of products A, B and C


(A)


(B)

$CH_3CH_2CH_2CH_2Br$

(C)

The mixture consists of

(A) A and B as major and C as minor products

(B) B as major, A and C as minor products

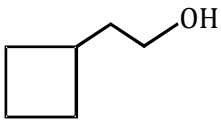
(C) B as minor, A and C as major products

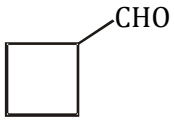
(D) A and B as minor and C as major products

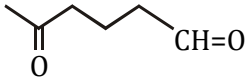
CHC003

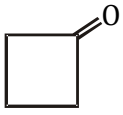
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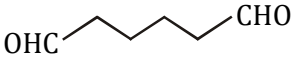
Ozonolysis of alkenes (Reductive and oxidative)

8.  $\xrightarrow[\Delta]{H^{\oplus}}$ A $\xrightarrow[(ii)Zn/H_2O]{(i)\ O_3}$ Product(s). Product is/are?

(A)  + HCHO

(B) 

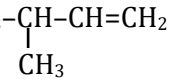
(C)  + CH_3CHO

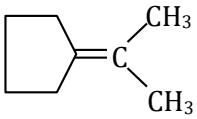
(D) 

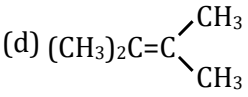
CHC179

9. Which of the following alkenes on reductive ozonolysis give ketones only?

(a) $CH_3–CH=CH–CH_3$

(b) 

(c) 

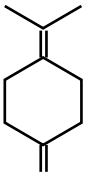
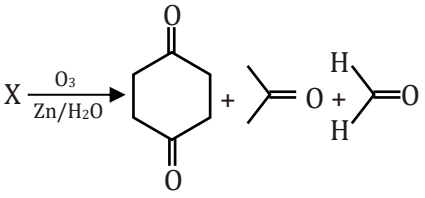
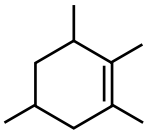
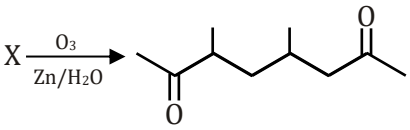
(d) 

(A) a and b (B) b and c (C) b and d (D) c and d

CHC182

For More Join: @IITJEE_Advanced

10. Match the Column :

List-I		List-II	
(A)	$X \xrightarrow[Zn/H_2O]{O_3} CH_3-CH_2CHO$	(P)	
(B)		(Q)	$Ph-CH=CH-CH_3$
(C)	$X \xrightarrow[Zn/H_2O]{O_3} PhCHO + CH_3CHO$	(R)	
(D)		(S)	$CH_3-CH_2-CH=CH-CH_2-CH_3$

CHC306

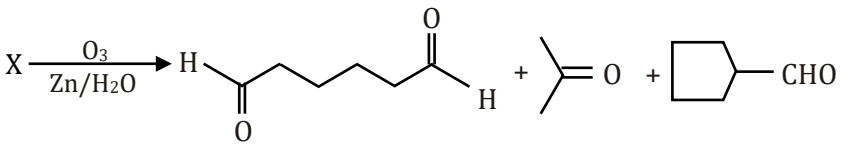
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11. $CH_3-CH=CH-\underset{\underset{H}{\vdots}}{\overset{\overset{CH_3}{|}}{C}}-CH=CH-CH_3 \xrightarrow[Zn+H_2O]{O_3}$ products ?

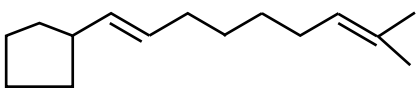
How many optically active product possible by above compound?

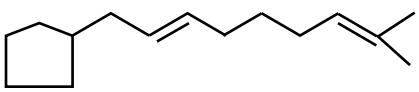
(A) 2 (B) 3 (C) 0 (D) 1

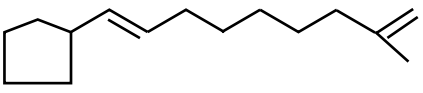
CHC307

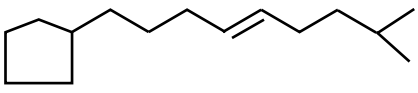
12. 

reactant 'X' is :

(A) 

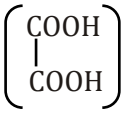
(B) 

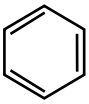
(C) 

(D) 

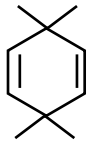
CHC308

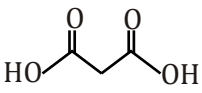
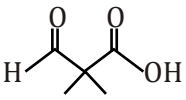
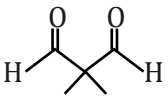
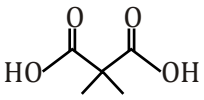
For More Join: @IITJEE_Advanced

13. Compound among following produces only oxalic acid  as major product on reaction with O_3/H_2O -

(A)  (B)  (C)  (D) None of these

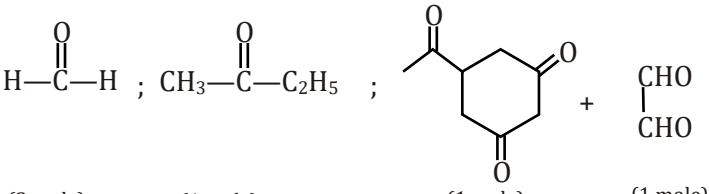
CHC309

14.  $\xrightarrow[2.H_2O]{1.O_3}$ Product:

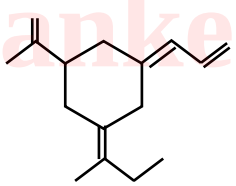
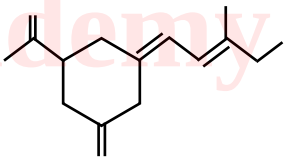
(A)  (B)  (C)  (D) 

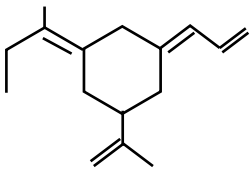
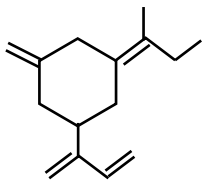
CHC310

15. By which compound's reductive ozonolysis the following products are obtained



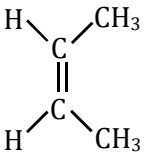
(2 mole) (1 mole) (1 mole) (1 mole)

(A)  (B) 

(C)  (D) 

CHC311

Oxidation of Alkenes (Baeyer's Reagent, OsO_4, Hot $KMnO_4$, Acidic $KMnO_4$, Peroxy Acid followed by hydrolysis, Ag/O_2 followed by hydrolysis)

16.  $\xrightarrow{\text{alkaline } KMnO_4}$ A, which is true about this reaction?

(A) A is meso 2, 3–butanediol formed by syn addition
(B) A is meso 2, 3–butanediol formed by anti addition
(C) A is a racemic mixture of d and l, 2, 3–butanediol formed by anti addition
(D) A is a racemic mixture of d and l, 2,3–butanediol formed by syn addition

CHC198

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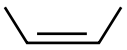
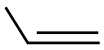
17. An alkene on treating with hot acidified $KMnO_4$ gives 4 - oxopentanoic acid. The alkene is

(A) Pentene (B) 2–Pentene

(C) 1–Methyl cyclobutene (D) 1, 2– Dimethyl cyclopropane

CHC200

18. Among the following which correctly follow the rate of epoxidation for the alkene :-

(i)  (ii)  (iii)  (iv) 

(A) (i) > (ii) > (iii) > (iv) (B) (i) > (iii) > (iv) > (ii)

(C) (iv) > (iii) > (ii) > (i) (D) (iv) > (iii) > (i) > (ii)

CHC203

Comprehension (Q.19 to 21)

From molecular formula C_9H_{10}, Total 4 Structural isomers [having monosubstituted benzene ring] are possible. Some structural isomer of molecular formula C_9H_{10} decolourises Bayer's Reagent and Br_2/CCl_4. Bayer's test is also test of unsaturation and it is positive for alkenes and alkynes. This test is negative for cyclopropane and benzene. Alkene and alkynes decolourize pink colour of Bayer's reagent (cold dilute alkaline solution of 1% $KMnO_4$).

19. How many of these structural isomers give Bayer's test of unsaturation ?

(A) 2 (B) 4 (C) 3 (D) 1

CHC312

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20. How many of these structural isomer can show Geometrical isomerism ?

(A) 2 (B) 1 (C) 3 (D) 4

CHC313

21. How many structural isomers give same product on catalytic hydrogenation at room temperature.

(A) 1 (B) 2 (C) 3 (D) 4

CHC314

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	A	AB	B	C	A	B	D	(A → S ; B → P ; C → Q ; D → R)
Question	11	12	13	14	15	16	17	18	19	20
Answer	C	A	B	D	ABC	A	C	D	C	B
Question	21									
Answer	B									

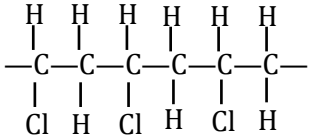
For More Join: @IITJEE_Advanced

Regular Analysis Through Continuous Exercise

Hydrocarbons : RACE-04

Polymerisation of alkene

1. The polymer shown is formed by the polymerisation of alkene.



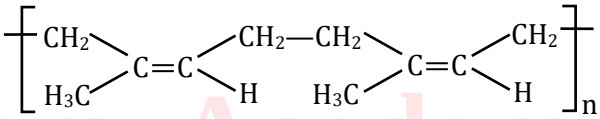
What is the molecular formula of the alkene which forms the polymer?

(A) $C_2H_2Cl_2$ (B) C_2H_3Cl

(C) C_3H_5Cl (D) C_3H_7Cl

CHC211

2. Natural rubber is given as follows



What products would be obtained on ozonolysis of its monomeric unit ?

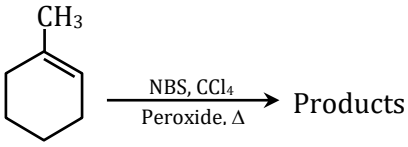
(A) Acetaldehyde, propanal (B) Formaldehyde, 2-ketopropanal

(C) Acetone, ethanal (D) Methanol, propanol

CHC213

Allylic substitution (NBS/Br_2)

3. Find out the total no. of products (including stereo) in the given reaction :



(A) 8 (B) 9 (C) 10 (D) 11

CHC183

Chemical reaction of alkyne ; Dimer, Trimer, Tetramer of acetylene

4. $\underset{\text{(Calcium carbide)}}{CaC_2} \xrightarrow{H_2O} (A) \xrightarrow{\text{Red hot Fe tube}} (B)$, Product (B) of the reaction is

(A) Toluene (B) Ethyl-benzene

(C) Benzene (D) Butyne

CHC232

For More Join: @IITJEE_Advanced

Polymerisation of alkyne

5. Propyne on cyclic polymerisation gives :

(A) Mesitylene (B) Benzene

(C) Ethyl benzene (D) Propyl benzene

CHC228

Chemical reactions of benzene ; Electrophilic Aromatic Substitution (EAS) of Benzene (Halogenation, Sulphonation, Nitration, Friedel Craft Alkylation, Friedel Craft Acylation)

6. In the sulphonation, acetylation and formylation of benzene the group of effective electrophiles would be :

(A) $SO_3^{\oplus}, CH_3C \equiv \overset{\oplus}{O}, \overset{\oplus}{H}CO$ (B) $SO_3, CH_3\overset{\oplus}{C}O, \overset{\oplus}{H}CO$

(C) SO_3, CH_3CHO, CO + HCl (D) HSO_3, CH_3CO, HCO

CHC241

7. Which of the following is most reactive towards sulphonation?

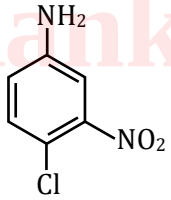
(A) m-Xylene (B) o-Xylene

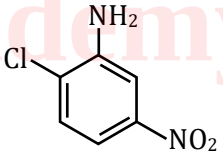
(C) Toluene (D) p-Xylene

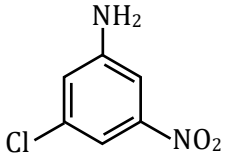
CHC244

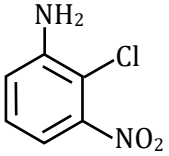
8. If meta-nitroaniline is chlorinated, the major product is:

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(A) 

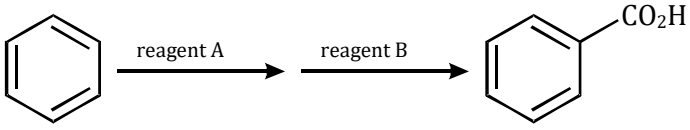
(B) 

(C) 

(D) 

CHC245

9. What reagents (conditions) are needed for the following reaction?



(A) reagent A : $CH_3COCl/AlCl_3$; reagent B : Sn/HCl, heat

(B) reagent A : $CH_3CH_2Cl/AlCl_3$; reagent B : $KMnO_4$, heat

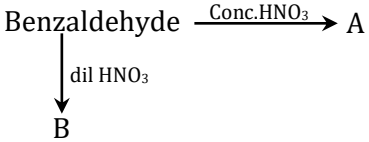
(C) reagent A : $CH_3COCl/AlCl_3$; reagent B : $H_2N–NH_2/KOH$, H_2O, heat

(D) reagent A : HNO_3, H_2SO_4; reagent B : Sn/HCl, heat

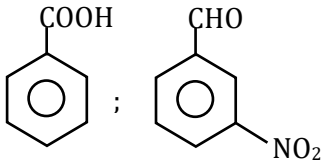
CHC249

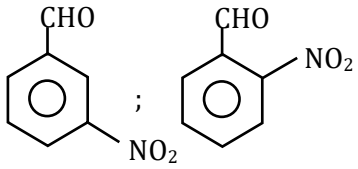
For More Join: @IITJEE_Advanced

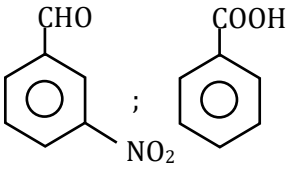
10. The major product of the following reactions are given as respectively

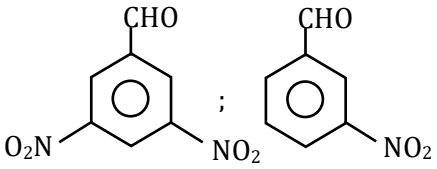


Identify A and B.

(A) 

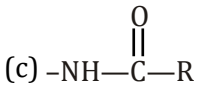
(B) 

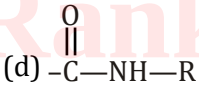
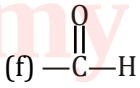
(C) 

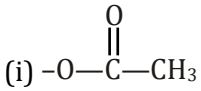
(D) 

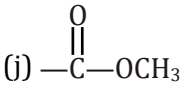
CHC251

11. Total number of atoms/groups, when attached on benzene are ortho/para directing & activating in nature towards electrophilic substitution are.

(a) –Me (b) – F (c) 

(d)  (e) $-C \equiv N$ (f) 

(g) $-NO_2$ (h) $-CF_3$ (i) 

(j) 

CHC315

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	B	B	C	A	B	A	B	B	C
Question	11									
Answer	3									

For More Join: @IITJEE_Advanced